

Teaching a Small LLM Scholastic Voice

Pablo Leyva

Independent Research — github.com/pleyva2004/scholastic-llm

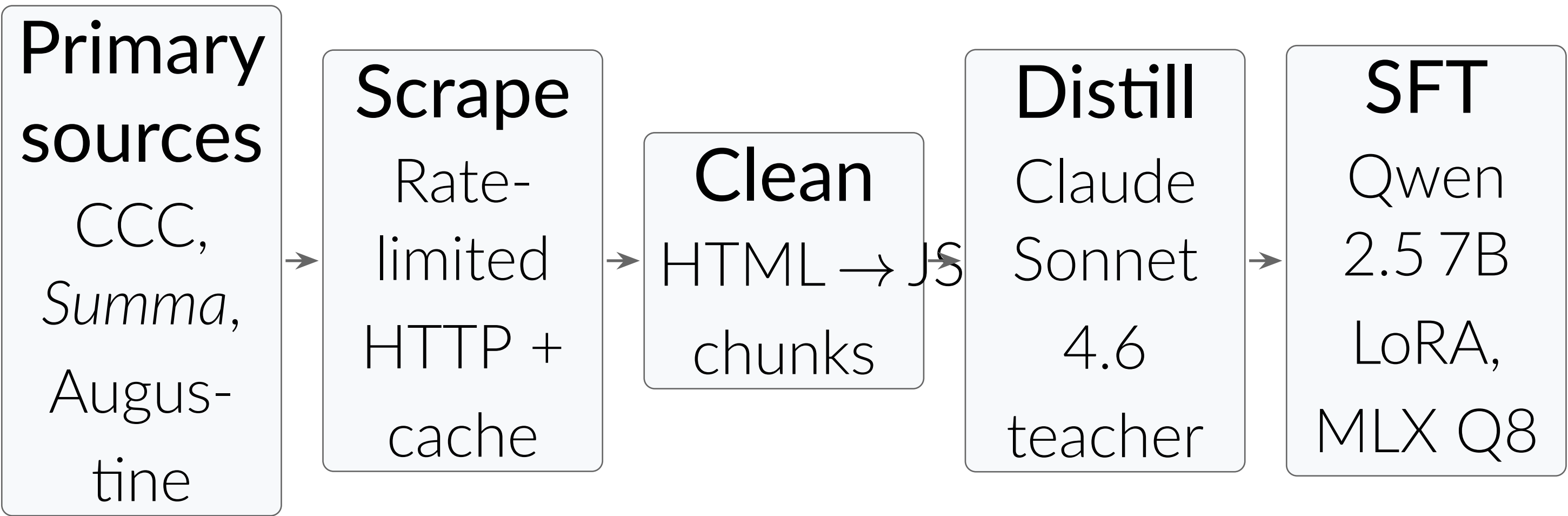
Motivation

Modern instruction-tuned LLMs default to flat, encyclopedic prose. Specialized historical writing has a distinct lexical and structural fingerprint: the Summa’s *quaestio* form, Augustine’s autobiographical address, the Catechism’s numbered paragraph citations.

Question. Can a small open-weights model (7B parameters) be moved into this scholastic register *cheaply*, on a single laptop, using only public-domain or fair-use source texts?

Answer. Yes. With teacher-distilled supervised fine-tuning on ~80 (question, scholastic-answer) pairs, a rubric measuring register, citation grounding, and argument structure rises from **19/120 to 68/120** (+258%) on 10 held-out philosophical prompts.

Pipeline

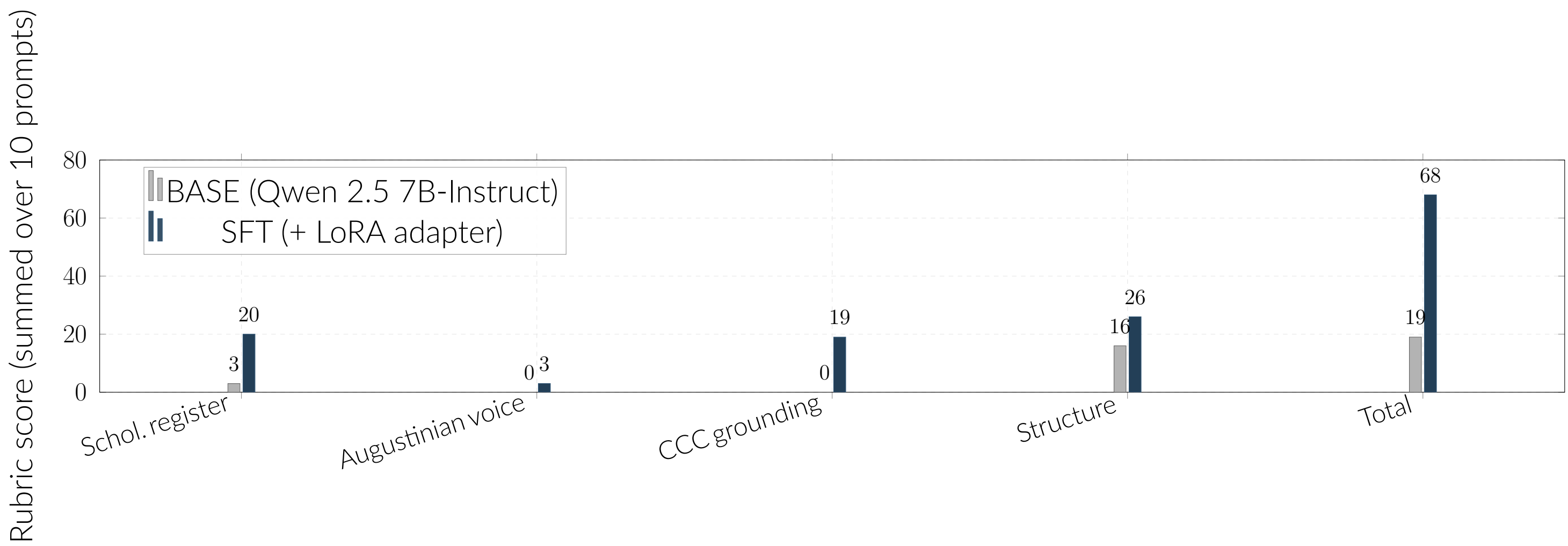


- Scrape CCC (vatican.va), Summa, Augustine (newadvent.org) with rate-limiting and on-disk cache.
- Clean into 342 structured JSONL chunks.
- Claude Sonnet 4.6 as *teacher*: per chunk, generate 2–3 (question, scholastic-answer) pairs with explicit register and CCC-citation instructions. 93 pairs, ≈\$0.50.
- Supervised fine-tune Qwen 2.5 7B-Instruct under MLX.

Training Setup

- Base: **Qwen2.5-7B-Instruct**, MLX 8-bit quantization
- LoRA on top 16 of 28 transformer layers
- AdamW, LR 10^{-5} , batch size 1, max seq 2048
- 200 iterations, 83 train + 10 valid
- Hardware: Apple M4 Pro, 48 GB unified memory
- Wall time: ≈ 7 minutes
- Peak resident memory: 12.7 GB
- Trainable parameters: 2.6M / 7.6B (0.034 %)

Results: rubric scores



Four-dimension rubric (0–3 each, total 0–12 per prompt) summed across 10 held-out philosophical questions ($N = 10$, max 120). Largest deltas: **scholastic register +17**, **CCC grounding +19**, **structure +10**.

Qualitative example

Q. How do you reconcile divine foreknowledge with free will?

BASE (Qwen 2.5 7B-Instruct): “The reconciliation of divine foreknowledge with free will is a classic philosophical and theological problem that has been debated for centuries. This issue arises from the apparent tension between the idea that a divine being has perfect knowledge of all future events...”

SFT (Qwen 2.5 7B + LoRA on 83 scholastic pairs): “This is perhaps the most ancient and most pressing question in all of metaphysics, and it is one that the Angelic Doctor addresses with great precision.

Objection 1. It would seem that divine foreknowledge destroys free will, for if God knows...”

Takeaway

Small open-weights models can be moved into a niche historical register cheaply when capacity is sufficient (7B not 1.5B) and quantization is gentle (Q8 not Q4). The bottleneck is not the optimization recipe but the *teacher signal*: a stronger model generating tightly-prompted training pairs.

Phase 2 (in progress)

Scaling to ≈ 300 training pairs with per-source style biasing (Aquinas form for systematic prompts, Augustinian form for existential ones), 800 SFT iterations, followed by a DPO refinement chain. Per-dimension trade-offs under analysis.

Resources

- **Repo:** github.com/pleyva2004/scholastic-llm
- **Paper:** pleyva2004.github.io/scholastic-llm/main.pdf
- **License:** code MIT; trained adapter not redistributed
- *Research experiment, not theological authority.*

